

Vid sand afl 2

Opgave 1.7

Betragt $(\Omega, \mathcal{F}, P)$ og $X = (X_1, \dots, X_d), Y = (Y_1, \dots, Y_m)$ være stokastiske vektorer derpå

Betragt også det "normale" indre produkt $\langle \cdot, \cdot \rangle_d, \langle \cdot, \cdot \rangle_m$

a) Antag $d = m$ vis $X \sim Y \Leftrightarrow \langle t, X \rangle_d \sim \langle t, Y \rangle_d \forall t \in \mathbb{R}^d$

Husk $X \sim Y \Leftrightarrow \phi_X(s) = \phi_Y(s)$ for alle passende t 'er

" $\Rightarrow$ " lad $s \in \mathbb{R}$ lad $t \in \mathbb{R}^d$ sæt $k = s \cdot t \in \mathbb{R}^d$

$$\begin{aligned} \phi_X(k) &= \phi_Y(k) \\ \mathbb{E}[e^{i\langle k, X \rangle_d}] &= \phi_X(k) = \phi_Y(k) = \mathbb{E}[e^{i\langle k, Y \rangle_d}] \\ \phi_{\langle t, X \rangle_d}(s) &= \mathbb{E}[e^{i\langle k, X \rangle_d}] = \phi_X(k) = \phi_Y(k) = \mathbb{E}[e^{i\langle k, Y \rangle_d}] = \phi_{\langle t, Y \rangle_d}(s) \\ &\Rightarrow \phi_{\langle t, X \rangle_d}(s) = \phi_{\langle t, Y \rangle_d}(s) \forall s \in \mathbb{R} \forall t \in \mathbb{R}^d \\ &\Rightarrow \langle t, X \rangle_d \sim \langle t, Y \rangle_d \forall t \in \mathbb{R}^d \end{aligned}$$

" $\Leftarrow$ " lad $t \in \mathbb{R}^d$

$$\begin{aligned} \langle t, X \rangle_d &\sim \langle t, Y \rangle_d \\ &\Rightarrow \mathbb{E}[e^{i \cdot k \cdot \langle t, X \rangle_d}] = \mathbb{E}[e^{i \cdot k \cdot \langle t, Y \rangle_d}] \forall k \in \mathbb{R} \end{aligned}$$

Specielt

$$\begin{aligned} \phi_X(t) &= \mathbb{E}[e^{i \cdot \langle t, X \rangle_d}] = \mathbb{E}[e^{i \langle t, Y \rangle_d}] = \phi_Y(t) \\ &\Rightarrow X \sim Y \end{aligned}$$

b) Vis $X \perp Y \Leftrightarrow \langle s, X \rangle_d \perp \langle t, Y \rangle_m \forall s \in \mathbb{R}^d, t \in \mathbb{R}^m$

husk

$$X \perp Y \Leftrightarrow \phi_{X,Y}(s, t) = \phi_X(s) \cdot \phi_Y(t) \forall s \in \mathbb{R}^d, t \in \mathbb{R}^m$$

" $\Rightarrow$ "

$\langle s, \cdot \rangle_d, \langle t, \cdot \rangle_m$ er lineære transformationer $\forall s \in \mathbb{R}^d, t \in \mathbb{R}^m$ som er målelige derved har vi at siden $X \perp Y$

$$\langle s, X \rangle_d \perp \langle t, Y \rangle_m$$

" $\Leftarrow$ "

Lad $s \in \mathbb{R}^d, t \in \mathbb{R}^m$

$$\phi_{X,Y}(s, t) = \mathbb{E}[e^{i(\langle s, X \rangle_d + \langle t, Y \rangle_m)}] = \mathbb{E}[e^{i\langle s, X \rangle_d} \cdot e^{i\langle t, Y \rangle_m}]$$

Da

$\langle s, X \rangle_d \perp \langle t, Y \rangle_m$ har vi igen da $e^{i(\cdot)}$ er en målelig funktion

Da giver uafh

$$= \mathbb{E}[e^{i\langle s, X \rangle_d}] \cdot \mathbb{E}[e^{i\langle t, Y \rangle_m}] = \phi_X(s) \cdot \phi_Y(t)$$

Derved $X \perp Y$